Artistic Value Model (AVM)

Purpose: a single, transparent formula that turns an artist's recorded music into an *artistic value* score, built from four parts: Vocal, Writing, Song as a Whole, Composition & Arrangement. Audience: data scientists and ML engineers. Supersedes the earlier "Artistic Analysis" write-up.

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1. The Formula at a Glance

Step 1: every metric becomes a sub-score (0–1)
(Section 2)
Step 2: sub-scores combine into four pillar scores (0–1) ,
summed only over sub-scores that could be measured reliably
Step 3: pillars combine into the Track Artistic Value (0–100)

```
None
TAV = 100 × ( W_V·P_V + W_W·P_W + W_S·P_S + W_C·P_C ) + PB
```

Pillar	Symbol	Default weight W
Vocal ability	P_V	0.30
Writing (lyrics and concept)	P_W	0.30
Song as a whole	P_S	0.20
Composition & arrangement	P_C	0.20

= production bonus, bonus-only: .
Production can add up to 3 points and can never subtract. Low production never lowers TAV (Section 6).
Step 4: tracks combine into the Artist Artistic Value (0–100)

```
None
AAV = 0.50·Median(TAV) + 0.20·P75(TAV) + 0.15·Consistency +
0.15·Trajectory
Consistency = 100 - 2 × StdDev(TAV)          (floored at 0)
Trajectory  = 50 + 10 × (slope of TAV per release, last 5 tracks)
(clipped to 0–100)
```

Weights are versioned configuration, never hard-coded. They start as expert-agreed defaults and are later learned from outcomes (Section 9). Genre profiles adjust them (Section 7).

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2. How a Metric Becomes a Sub-Score

Raw metrics (cents of pitch error, rhyme density, etc.) are not comparable. Each goes through calibration:

1. Craft metrics with a clear direction (e.g. pitch error): fit a monotonic (isotonic) curve from the metric to expert ratings. Example: 10 cents average error → 0.9, 40 cents → 0.4.
2. Norm-dependent metrics (e.g. repetition, energy shape, genre fit): score against the peer group (same genre, country, era) with a "sweet spot" function:
   , where is the optimal value learned from successful tracks in that peer group and σ the tolerance. Too little *and* too much both score lower.
3. Everything gets a confidence value (0–1) and evidence (timestamps, lyric lines). Low confidence widens the final score range; it does not silently change the score.

Peer groups need at least 30 tracks; otherwise use a broader group and set the flag.

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3. Pillar V: Vocal Ability (P_V)

First, build a clear picture of the voice. All vocal analysis runs on the isolated vocal stem (Demucs). Live footage is used where available, because it is the most honest evidence.

#	Sub-score	What is measured	How	Default w
V1	Pitch accuracy / key-holding	Average and distribution of deviation in cents from the intended note and scale; time spent in key	F0 tracking (CREPE or pYIN) → snap to detected key/scale → deviation per note	0.25
V2	Note stability	Steadiness on held notes, vibrato rate/depth, wobble, pitch drift at note ends	F0 variance on sustained segments; vibrato analysis	0.15
V3	Tone / voice quality	Richness, clarity, breathiness, strain, consistency of timbre	Spectral features (MFCC, formants, harmonics-to-noise ratio), singing-quality embeddings, expert-trained regressor	0.20
V4	Range	Lowest and highest *usable* notes, usable range in semitones, comfortable tessitura	Voiced F0 5th–95th percentiles, only counting notes held stably and on-pitch; report lowest/highest note names	0.15
V5	Expression and dynamics	Loudness variation, phrasing, emotional delivery	Loudness contour on vocals; emotion model	0.15
V6	Timing and phrasing	Sits with the beat, rhythmic precision, phrase length and breath control	Onset alignment to beat grid; phrase-length distribution	0.10

Range scoring: , not raw semitones. A rapper is not penalised for narrow range because the genre profile lowers its weight (Section 7). Always display the raw lowest and highest notes too.

Autotune and processing rule (critical):

1. Detect processing: hard pitch-snapping signature (unnaturally quantised F0 with near-zero micro-variation, step changes at note onsets), formant shifting, heavy effects → set and a (0–1).
2. Autotune is reported, not punished. It is a fact surfaced to A&R.
3. When is high, V1 and V2 become "unverifiable" and are removed from the weighted average (Step 2 renormalises over what remains) with confidence reduced. Where raw or live vocals exist, compute V1/V2 from those instead.
4. Output so A&R knows whether the pitch scores reflect the singer or the software.

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4. Pillar W: Writing (P_W)

Run on the transcribed lyrics (language-identified per line, since songs code-switch). Raters and tools are language-specific (English, Swahili, Sheng, Pidgin, Yoruba, Lingala etc.).

#	Sub-score	What is measured	How	Default w
W1	Coherence	Do the lines hang together; clear theme; narrative or emotional arc; no random topic jumps	Semantic similarity between consecutive lines and sections; topic drift; LLM rubric score calibrated to raters	0.20
W2	Imagery	Concrete, sensory, specific language; metaphor and simile that land	Concreteness and sensory lexicons; figurative-language detector; LLM rubric score	0.20
W3	Repetition balance	Repetition that serves the song (hook, refrain) vs lazy looping	Repetition ratio r vs peer-group optimum r^* using the sweet-spot function (Section 2)	0.10
W4	Memorability	Could a listener recall and sing it	Hook line length (short is better), hook repetition, simple vocabulary in the hook, rhythmic catchiness, singability	0.15
W5	Concept originality	Fresh angle on the subject vs a well-worn one	Embedding distance of the song's theme/concept from the peer group (sweet-spot function: distinct but understandable)	0.15

W6	Cliché avoidance	Stock phrases and overused rhymes	Cliché-phrase dictionary (per language) + embedding matches to a stock-lyrics corpus; `s = 1 − cliché_rate_normalised`	0.10
W7	Sound craft	Rhyme scheme, internal and multisyllabic rhyme, alliteration, meter and prosody (stress lands on beats)	Grapheme-to-phoneme → rhyme detection per language; syllable-to-beat alignment	0.10

Other written-art criteria (voice, point of view, economy, emotional range) can be added as W8+ once raters can score them reliably.
Key rule: rhyme, cliché and repetition norms differ by language and genre; all are calibrated per peer group. Never apply English-only resources to other languages.

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5. Pillar S: The Song as a Whole (P_S)

Pillars V and W judge elements. This pillar judges whether the elements work together and whether the song's overall shape is a proven one.

#	Sub-score	What is measured	How	Default w
S1	Energy curve and structure fit	Shape of the energy curve; does the structure follow a form known to work in the genre	Energy curve (loudness, onset density, spectral flux, bass weight, brightness, layering; 1 point per second). Metrics: time to first peak, builds, drops, section lift, shape label. Structure fit = similarity (curve and section layout) to the median curves of successful tracks in the peer group	0.35
S2	Cohesion	Do lyrics, melody, vocals and arrangement point the same way	Mood match between lyrics and music (valence/arousal alignment); melody–lyric stress alignment; vocal-to-beat fit; section-level agreement of energy with lyrical intensity	0.35
S3	Holistic quality	The gestalt, "more than the sum of its parts"	Model trained on expert whole-song ratings, using all pillar features plus cross-features (e.g. strong hook × strong vocal on the hook)	0.30

Production value is measured here but kept outside P_S: , scored 0–100 and reported separately as the Production Index. It feeds only the capped bonus (Section 1) and the "surface" rule (Section 6).

Energy-curve reference code already exists ( ) and implements S1's curve and metrics.

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6. Production Never Blocks Discovery

1. Production is not in the weighted sum; it can only add up to +3 points.
2. Vocal, writing and composition features are extracted from separated stems and judged for what they are, even if the mix is poor. Low-quality audio lowers *confidence* (wider range), not the score.
3. Surface rule: an artist is surfaced as "High potential, low production" when and . These are shown with their own tag; for A&R, this is often the best signal, since production can be bought.
4. The Production Index is reported as a polish headroom measure for planning budget.

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7. Pillar C: Composition & Arrangement (P_C)

#	Sub-score	What is measured	How	Default w
C1	Melody	Range, contour, interval variety, tension and resolution, singability	Pitch tracking of the lead melody; contour/interval statistics; resolution to stable scale degrees	0.30
C2	Harmony	Chord vocabulary, progression quality, key changes, melody–chord fit	Chord recognition; progression n-gram statistics vs peer-group norms; melody-vs-chord compatibility	0.25
C3	Rhythm and groove	Groove, syncopation, variation, micro-timing	Onset/beat tracking; syncopation index; swing and micro-timing	0.15
C4	Arrangement	Layering, instrument roles, contrast between sections, use of space	Stem activity maps; per-section texture change; frequency-space occupancy	0.20

C5	Genre mastery and creative deviation	Does the artist know the genre's rules, and break them well	Distance from genre centroid in audio-embedding space (MERT/CLAP/MusicFM); sweet-spot function with `d*` learned from successful tracks; fusion coherence for multi-genre tracks	0.10

Caveat: complexity is not quality (many hits are simple). C-scores use the sweet-spot approach and are validated against outcomes (Section 9) before being trusted.

Genre profiles (adjust weights; examples to be tuned with experts)

Profile	Changes
Rap / hip-hop	P_V: V1, V2, V4 weights cut to 0.05 each; V6 (flow/timing) raised to 0.40, V5 to 0.25; P_W weight raised to 0.40, P_V lowered to 0.20
Afrobeats / Amapiano	Raise C3 (rhythm/groove) to 0.30; raise S1 (energy structure); lower W3 repetition penalty (repetition is a stylistic feature)
Gospel / soul / R&B	Raise V3, V4, V5; raise P_V to 0.35
Folk / singer-songwriter	Raise P_W to 0.40; lower C4 arrangement and Production

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8. Output Record

```
None
entity: track | artist, id, as_of_time, model_version,
weights_version, genre_profile
TAV (0–100) with range [low, high] from confidence
pillars{ V, W, S, C } each 0–1 with confidence
sub_scores{ V1..V6, W1..W7, S1..S3, C1..C5 } each with value,
confidence, evidence[]
vocal_profile{ lowest_note, highest_note, usable_range_semitones,
key, processing_level, intrinsic_pitch_confidence }
production_index, PB
flags[]: processed_vocals, compressed_master, low_quality_audio,
lyrics_uncertain, broad_cohort,
         insufficient_evidence, high_potential_low_production
artist rollup (AAV, median, P75, consistency, trajectory) when ≥ 3
tracks
```

If a sub-score cannot be measured, it is omitted (not zeroed); if a whole pillar cannot be measured, output for the track.
Worked example (simplified) Pillars: V = 0.72, W = 0.65, S = 0.70, C = 0.60.

Production Index 40 → PB = 0 → TAV = 67.1. If the peer-group 75th percentile is 62, the track is tagged .

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9. Training, Calibration and Proof

1. Rater programme: 3–5 qualified raters (musicians, producers, A&R, native-language lyricists) score 1,000–3,000 songs across the sub-score rubrics (1–5), plus pairwise comparisons for hard items and an overall whole-song rating. Measure inter-rater agreement (Krippendorff's alpha / ICC); items raters can't agree on are marked subjective and carry lower confidence.
2. Metric models: per sub-score, start with interpretable models (isotonic calibration, ridge, gradient boosting) predicting the expert rating; add audio and text embeddings; calibrate LLM rubric scoring to raters per language. Use ensembles or conformal prediction for confidence.
3. Initial weights: elicited from an expert panel (e.g. pairwise comparison of pillars), recorded as version 1.
4. Learned weights: once outcome data exists (breakout, durable career, one-hit), fit pillar and sub-score weights per genre profile using survival analysis or gradient boosting, and compare with expert weights.
5. Predictive-value test: a point-in-time backtest comparing the career-outcome model with and without AVM features. If a pillar adds no lift, down-weight it. A score that experts like but that predicts nothing should not drive decisions.
6. Other checks: bias audit by genre, language, country and production budget; stability to re-encoding and trimming; golden-set regression tests on every release; A&R corrections logged as new labels.

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10. Build Order

Phase	Build	Result
1	Preprocessing (stems, lyrics, language ID, processing detection); V1–V4 vocal facts incl. range and autotune flag; W-lexical metrics (diversity, cliché, rhyme, repetition); energy curve; key/chords	Verifiable facts, e.g. "range F2–A4, 91% of held notes within 20 cents, autotune detected"
2	Rater programme; calibrate all sub-scores; peer-group normalisation; compute pillars and TAV (expert weights)	First TAV with confidence
3	W1/W2/W5 (coherence, imagery, originality); S2/S3 (cohesion, holistic); C5 genre mastery; genre profiles; AAV roll-up	Full four-pillar model

4	Outcome validation, learned weights, bias audit, "high potential, low production" tagging, A&R feedback loop	Evidence the formula predicts real careers

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11. Decisions Needed

1. Confirm the four pillar weights and the production bonus cap (default 3 points).
2. Starting genres, countries and languages.
3. The rater panel and how they are engaged.
4. Outcome definitions for validation (breakout, durable career).
5. Surfacing threshold (default: peer-group 75th percentile).